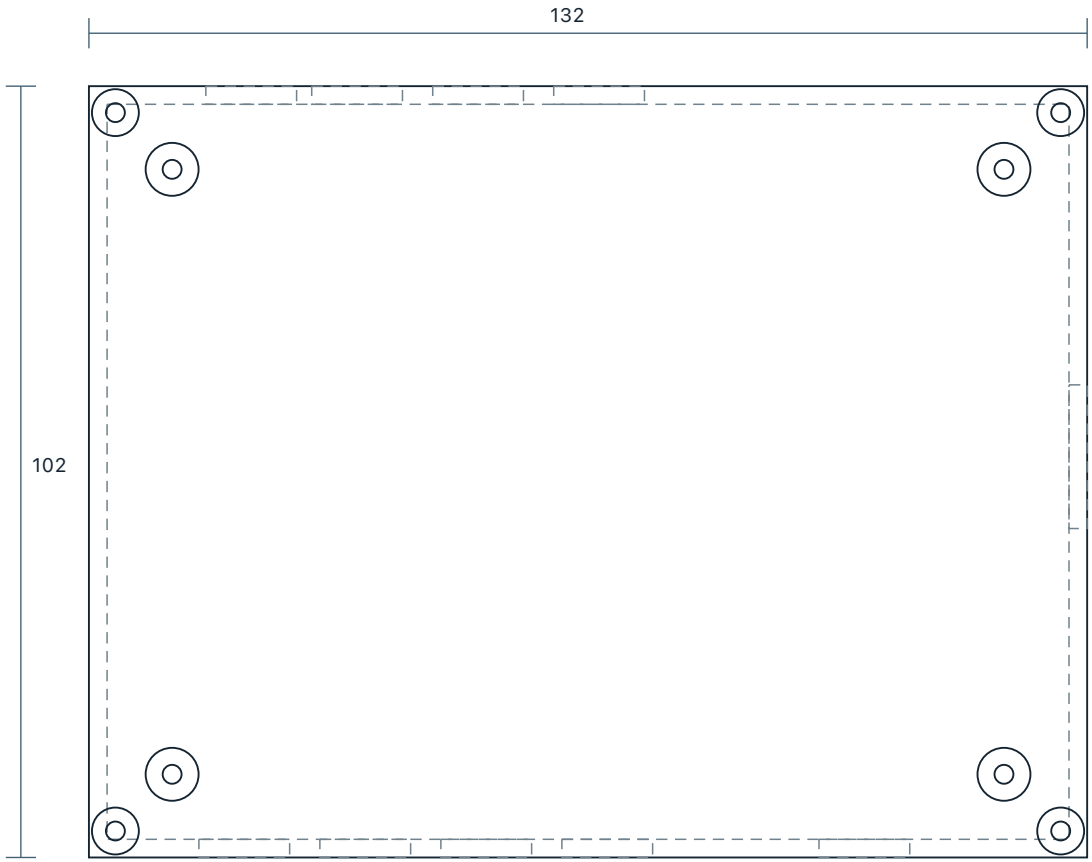
# tinyPLC A0 / BASE / TOP AND SIDE VIEWS

ENGINEERING DRAFT - mm - scale 1:1 at actual-size A4 printing - NOT A PCB FABRICATION RELEASE




TOP: XY datum = outside corner at left/top; Z datum = outside floor

CUTOUTS: 4 DI windows on Y=0 edge; 4 Q + power windows on Y=102 edge

See component placement/template for terminal centers. M3 pilots are tapped, not heat-set-insert sockets.

## SIDE / SECTION HEIGHTS




Floor: 2.4; walls: 2.4

PCB underside Z=8; top Z=9.6

PCB support: OD 7; pilot 2.5

Support centers: (11,11), (121,11),

(11,91), (121,91)

Lid towers: OD 6.2; pilot 2.5

Terminal windows: 12 W x 16 H

Window bottom Z=9; top Z=25

USB: 19 W x 10 H; bottom Z=7

USB Y span: 39.5 to 58.5

Pilot sizes require printer coupon.

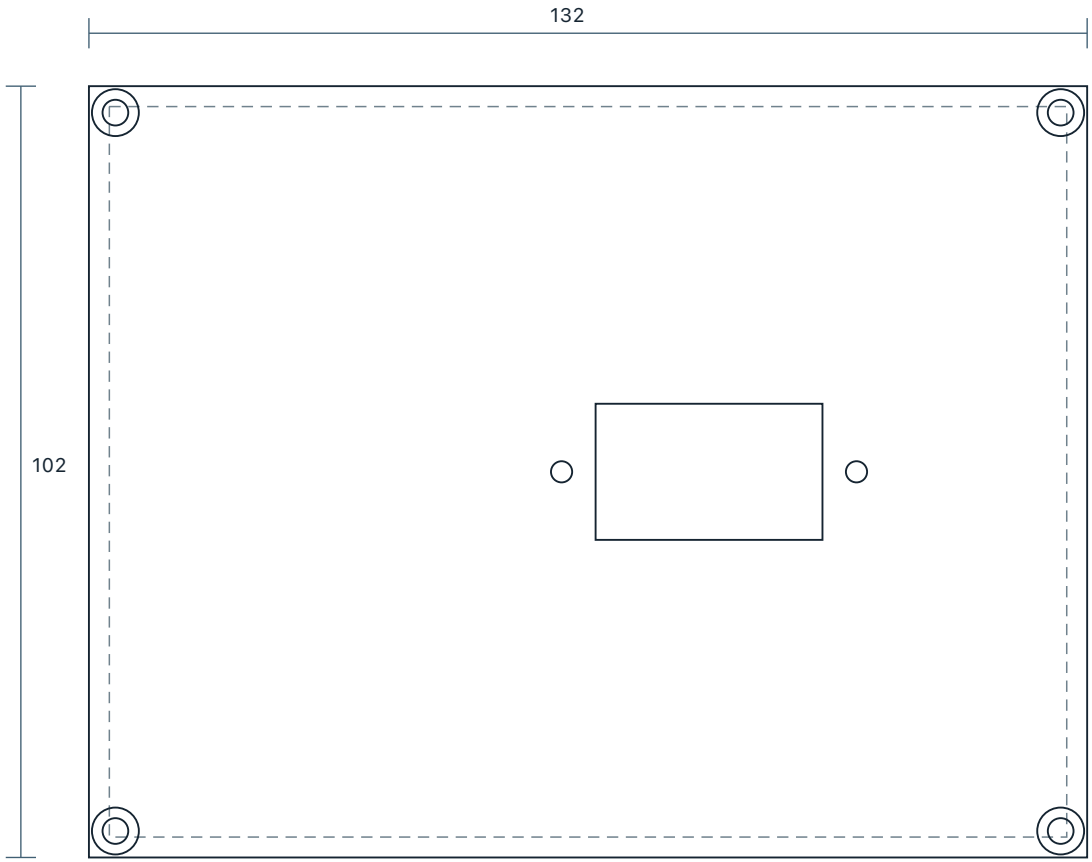


Check the scale bar. Print at 100%; disable fit-to-page. Native FreeCAD and parameters govern geometry.



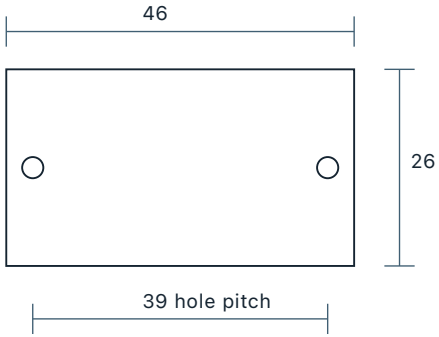
# tinyPLC A0 / LID AND DISPLAY BLANK

ENGINEERING DRAFT - mm - scale 1:1 at actual-size A4 printing - NOT A PCB FABRICATION RELEASE




TOP VIEW. Window center: X=82, Y=51. Window: 30 x 18.

DISPLAY BLANK




Lid plate: 2.4 thick

Locating lip: 1.8 below lid

Lip outer: 126.6 x 96.6

Lip wall: 1.6 nominal

Lip clearance: 0.3 per side

4 holes: dia 3.4 through

Counterbores: dia 6.2 x 1 deep

Blank: 2 thick, 2 holes dia 2.8

2 lid/blank holes: dia 2.8

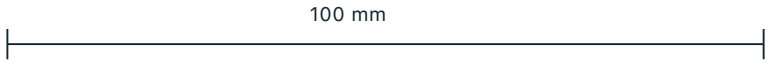
Blank fasteners: M2.5 x 8 + nuts

Lid screw centers: (3.5,3.5), (128.5,3.5), (3.5,98.5), (128.5,98.5). M3 x 8 candidate screws.

Display blank is a cover; no OLED bracket is defined. Lip has local tower clearances; inspect STEP/FreeCAD.

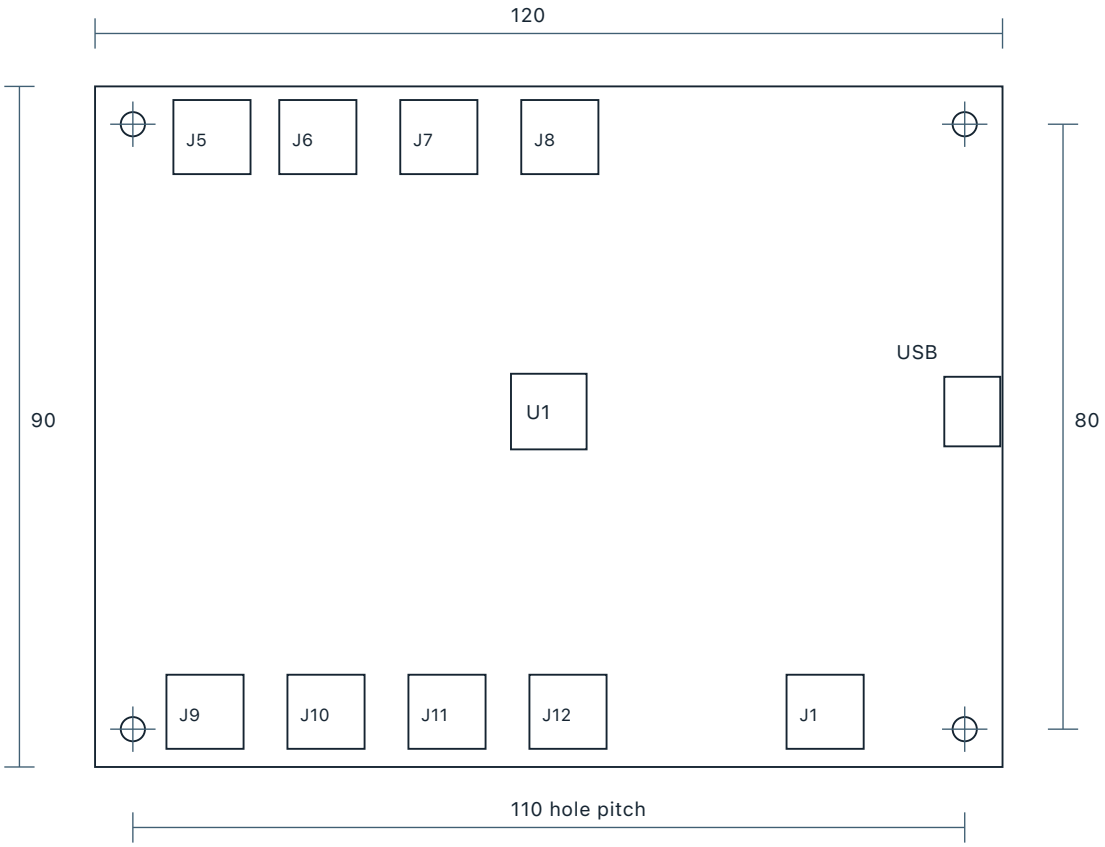


Check the scale bar. Print at 100%; disable fit-to-page. Native FreeCAD and parameters govern geometry.



# tinyPLC A0 / PCB FIT TEMPLATE / COMPONENT PLACEMENT

ENGINEERING DRAFT - mm - scale 1:1 at actual-size A4 printing - NOT A PCB FABRICATION RELEASE




Board: 120 x 90 x 1.6

Mounting holes: 4 x dia 3.2 NPTH

Centers relative to board datum:

(5,5), (115,5), (5,85), (115,85)

DI edge = Y=0 (top of this view)

Q/power edge = Y=90 (bottom)

J5 DI1 / J6 DI2 / J7 DI3 / J8 DI4

J9 Q1 / J10 Q2 / J11 Q3 / J12 Q4

J1 power / J13 USB-C

Board-to-case offset: +6,+6

Template can be cut from card.

THIS IS NOT A COPPER/PAD ARTWORK.

TOP VIEW - label side up. A0 footprint placements are provisional; use KiCad for all pads and pin numbers.

The mounting-hole and terminal envelope template is for mechanical fit only. The electronic PCB is unrouted.



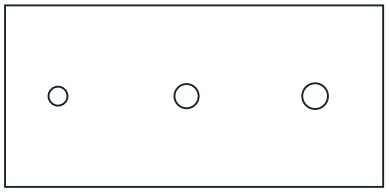
Check the scale bar. Print at 100%; disable fit-to-page. Native FreeCAD and parameters govern geometry.

100 mm

# tinyPLC A0 / PRINT COUPON AND ASSEMBLY STACK

ENGINEERING DRAFT - mm - scale 1:1 at actual-size A4 printing - NOT A PCB FABRICATION RELEASE

## 50 x 24 x 5 test coupon



2.5 pilot 3.2 3.4

Tap pilot and try actual screws before the full print.

Measure hole size, elephant foot, layer bonding and flatness.

## PRINT ORIENTATION

base-print.stl: floor down, open side up

lid-print.stl: outside face down, lip up

display-blank-print.stl: flat

fit-coupon-print.stl: flat

Import STL units as mm; do not rescale.

0.4 nozzle / 0.2 layer is a starting proposal.

Check bridges/supports in the slicer first.

Material and settings are not physically qualified.

## STACK / assembled Z coordinates:

0.0 outside base bottom

2.4 inside floor

8.0 PCB underside

9.6 PCB component side

23.4 terminal top (nominal envelope)

28.2 underside of lid locating lip

30.0 lid underside / tower top

32.4 lid outside face

34.4 blank outside face

Screw heads may project further.



Check the scale bar. Print at 100%; disable fit-to-page. Native FreeCAD and parameters govern geometry.

100 mm